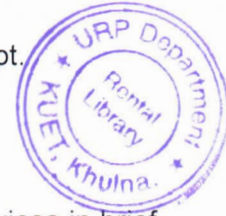


Khulna University of Engineering & Technology  
 Department of Urban and Regional Planning  
 BURP 4<sup>th</sup> Year 1<sup>st</sup> Term Regular Examination, 2021  
**URP 4111**  
 (Utility Services and Infrastructure Planning)

Full Marks: 210

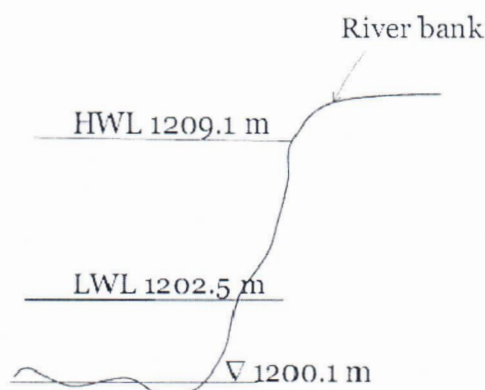
Time: 3 hrs

- N.B.** i) Figures in the right margin indicate full marks.  
 ii) Answer any **three** questions from each section in separate script.



**Section – A**

1. (a) Discuss site and community specific characteristics of utility services in brief. (10)  
 (b) Exemplify the components of Water Supply System with necessary illustrations. (15)  
 (c) Write a short note on “State monopoly of utility services”. (10)
2. (a) What is meant by inelasticity of water demand? Mention some measures of water demand management. (15)  
 (b) Write a short note on “Average water demand vs. specific water demand”. (10)  
 (c) In a residential area of 250,000 inhabitants, the specific water demand is estimated as 200 l/c/d, which includes leakage. Calculate the demand in 20 years’ time if the assumed annual demand growth is 2.86%. Compare the results by applying the linear and exponential models. (10)
3. (a) Illustrate any three from the following water intake structures with sketches: (15)
  - i. Floating Intakes
  - ii. Submerged Intakes
  - iii. Tower Intakes
  - iv. Shore intakes
  - v. Pier Intakes
- (b) Given the following information propose a suitable river intake (consider 2 circular sumps and a free board of 1m). Assume necessary data which are missing. (15)
  - Daily Demand: 6000 m<sup>3</sup>
  - Pump Capacity: 55 l/s (working 8.5 hours/day)



- (c) Write down the advantages of Rainwater Harvesting (RWH). (05)
4. (a) What are the basic rules of fire-fighting demand calculation and future water demand forecasting? (10)  
 (b) Estimate the fire flow rate and volume of water required to provide protection to a 15 storey non-combustible building with an effective floor area of 7000 square meter. (10)  
 (c) Describe different types of water distribution network with sketches. (15)

## **Section – B**

5. (a) What do you mean by soft and hard infrastructure? (05)  
(b) Mention the relations among infrastructure, economic growth and poverty reduction with the help of some major service sector. (10)  
(c) How do technology and fiscal constraints influence the decision of infrastructure investment? (10)  
(d) Differentiate between financing and funding. (10)
6. (a) What do you mean by spillover benefit? (05)  
(b) Describe the common sources of government funding in the context of Padma Bridge. (15)  
(c) How the price of infrastructure can be established? Consider the Shahid Haidar Park, Khulna for the purpose. (10)  
(d) Draw your concept about "Time Inconsistency" in infrastructure decision. (05)
7. (a) What are the non-political goals of infrastructure investment? (05)  
(b) How does infrastructure influence the supply and demand side production? (15)  
(c) How does infrastructure create the market failure? Describe with examples. (10)  
(d) Write down some benefits of Public-Private-Partnership. (05)
8. (a) Considering the newly applied "Electronic Toll Collection System in Padma Bridge"- describe the factors of technological application in infrastructure operation. (15)  
(b) Khulna-WASA is running "Water Supply Project" throughout the Khulna City. How will you define the infrastructure service delivery cost in here? (10)  
(c) Infrastructure decisions have some measures that go beyond- benefit/cost analysis. - Justify the statement for Padma Multipurpose Bridge Project. (10)

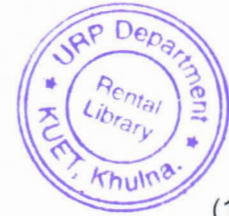
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Khulna University of Engineering & Technology  
Department of Urban and Regional Planning  
BURP 4<sup>th</sup> Year 1<sup>st</sup> Term Regular Examination, 2021  
**URP 4141**  
(Environmental Planning and Management)

Full Marks: 210

Time: 3 hrs

- N.B.** i) Figures in the right margin indicate full marks.  
ii) Answer any **three** questions from each section in separate script.



**Section – A**

1. (a) Draw your concept on Environment, Ecology and Ecosystem. (10)  
(b) Discuss the interrelationships among population, environment and development. (15)  
(c) Annotate the followings: (10)
  - i) Development Vs Sustainability
  - ii) Equity in opportunity and development
2. (a) What is meant by air pollution? Discuss various effects of air pollution on our health, environment and other wellbeing. (10)  
(b) How do the local meteorological and non-meteorological conditions affect the transport, dilution and ultimate effects of air pollutants? Explain. (15)  
(c) Discuss different schemes that are adopted globally to reduce carbon emission and to mitigate carbon footprint. (10)
3. (a) How does persuasive instrument of environmental management work? Discuss in brief. (10)  
(b) What do you mean by market-based instrument (MBI) of environmental management? Discuss in brief the important MBIs. (15)  
(c) Explain the ecological rating system for impact evaluation with an example. (10)
4. (a) What is tragedy of common? Discuss the characteristics of public goods. (08)  
(b) Draw your concept on environmental externalities. How these could be internalized? Explain with necessary example. (12)  
(c) Discuss the Hedonic pricing method and Contingent Valuation method of valuation of non-market [environmental] goods and services. (15)

**Section – B**

5. (a) Distinguish between SEA and EIA. (08)  
(b) Why EIA is necessary for implementation of any large scale project? Discuss in brief. (12)  
(c) What is meant by the Impact Statement? Discuss the main elements of an Impact Statement with necessary example. (20)
6. (a) List out the different methods of impact prediction. (08)  
(b) In which method of impact identification, the probability of occurrence is considered? Explain that with necessary example. (12)  
(c) Write down the different types of impact and their attributes that are considered in the EIA process. (15)
7. (a) EIA covers time frames. Explain the different time frames with necessary (15)

example.

- (b) What is sensitivity analysis in EIA? Illustrate with necessary example. (10)
  - (c) Discuss the uncertainties of impact prediction. (10)
8. (a) An industry produces 500 m<sup>3</sup>/day of organic waste with a BOD<sub>5</sub> of 537 g/l. If wastes with BOD<sub>5</sub> less than 300 mg/l are covered by basic sewer charges through taxes or as part of the water bill, what bi-annual surcharge this industry would pay if the operating cost for removing BOD<sub>5</sub> is \$0.25 per kg BOD<sub>5</sub> removed. (15)
- (b) How can you control the pollution on the basis of Economic Aspects? Discuss your idea. (12)
  - (c) Why public/stakeholder participation is necessary for EIA process? Discuss in brief. (08)

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Khulna University of Engineering & Technology  
Department of Urban and Regional Planning  
BURP 4<sup>th</sup>Year 1<sup>st</sup> Term Examination, 2021  
URP 4175  
(Resettlement Planning)

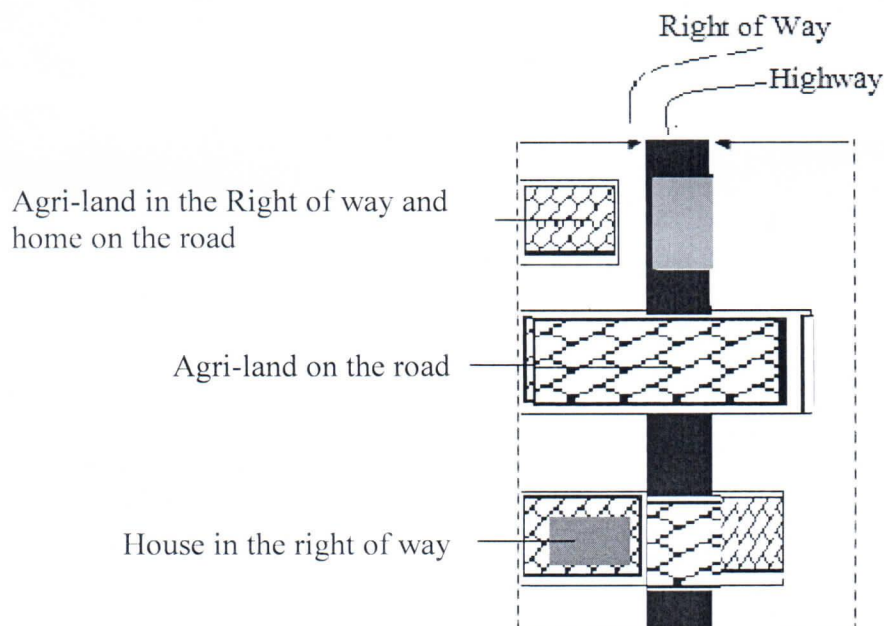
Full Marks: 210

Time: 3.00 hrs

- N.B. i) Answer any THREE questions from each section in separate scripts.  
ii) Figures in the right margin indicate full marks.

**Section – A**

1. (a) "Land acquisition is the biggest constraint for infrastructure projects in Bangladesh"-explain. (07)  
(b) Write short notes on the following: (16)
  - i. Displacement
  - ii. Resettlement
  - iii. Compensation
  - iv. Rehabilitation(c) Why and how displacement and resettlement is a global concern? Explain. (12)
2. (a) What is Resettlement Action Plan (RAP) and what are its major components? (20)  
Mention the steps associated with the Project Impact and Affected Population identification of RAP?  
(b) Following figure shows three possible displacement and resettlement. How would you consider compensation? (15)



3. (a) Explain in short, the Maslow's Hierarchy of Needs. Explain how poor resettlement planning can affect the Hierarchy of Needs and to what extent. (17)  
(b) Explain the key principles of resettlement planning. (08)  
(c) What are the conditions for 'land for land' and 'cash compensation'? Which one of the two compensation types is better and why? (10)
4. (a) Why World Bank Resettlement policy is a milestone document? Mention the basic principles of the WB Resettlement policy. (12)  
(b) Explain in short why Jamuna Bridge is a landmark project in Bangladesh? In reference to post-resettlement outcomes, what are the consequences on the affected community? (15)  
(c) Explain how racism can play a big role in unequal distribution of compensation. (08)

## **Section – B**

5. (a) What are the important features of 1982 Ordinance and 1989 Act? What is the procedure for award of compensation by the Deputy Commissioner? (15)  
(b) Address some of the loopholes of 1982 ARIPO Ordinance for finalizing land price. (10)  
(c) "The present resettlement policies are adequate."- Give your opinions for or against this statement. (10)
6. (a) "The eligibility criteria for benefit are itself a major shortcoming"- Explain with the help of a suitable example. (15)  
(b) Explain the issues in the Economics of Resettlement. (15)  
(c) What are the policy objectives of the World Bank on Involuntary Resettlement? (05)
7. (a) Why is economic analysis essential to Resettlement? (10)  
(b) Explain the limitations of Cost-Benefit Analysis for Involuntary Resettlement. (15)  
(c) According to the IRR (Impoverishment Risk and Reconstruction) model, mention some of the risks and their counterparts. (10)
8. (a) House components from any of the many acts, ordinances, and laws to create an ideal version of a resettlement policy in Bangladesh. (15)  
(b) Write a case study showing the lessons learned and difficulties during an Involuntary Resettlement. (10)  
(c) What are the advantages of integrating IRR model with current resettlement practices? How can they reduce the risk to displaced communities? (10)

Khulna University of Engineering & Technology  
Department of Urban and Regional Planning  
BURP 4<sup>th</sup> Year 1<sup>st</sup> term Examination, 2021  
**URP 4161**  
(Project Planning and Management)

Full Marks: 210

Time: 3.00hrs

- N.B.** i) Figures in the right margin indicate full marks.  
ii) Assume reasonable value for any missing data.  
iii) Answer any three questions from each section in separate script.



**Section – A**

- |   |                                                                                                                                                                                                 |    |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 1 | (a) Draw your concept on project life cycle.                                                                                                                                                    | 05 |
|   | (b) Illustrate – Vision, Mission, Policy, Strategy, Plan and Program with necessary example.                                                                                                    | 15 |
|   | (c) Discuss various plans that are required to prepare in project management cycle.                                                                                                             | 15 |
| 2 | (a) Clarify the terms : Baseline, Endline, Audit and Review.                                                                                                                                    | 08 |
|   | (b) Discuss various types of evaluation of projects with necessary example.                                                                                                                     | 15 |
|   | (c) Prepare a conflict management plan for a multi-stakeholder engaged small-scale surface irrigation project in Bangladesh.                                                                    | 12 |
| 3 | (a) Draw your concept on Logical Framework Approach ( LFA) of Project design                                                                                                                    | 08 |
|   | (b) Suppose Khulna shipyard road has to be reconstructed. Now, make a logical framework ( Log-frame) as part of Development Project Proposal ( DPP) for reconstruction of Khulna Shipyard Road. | 15 |
|   | (c) Prepare a hypothetical problem tree analysis of urban WASH problems.                                                                                                                        | 12 |
| 4 | (a) Draw your concept on RBM                                                                                                                                                                    | 05 |
|   | (b) Distinguish between monitoring and evaluation                                                                                                                                               | 10 |
|   | (c) Write a short note on : economic feasibility vs. financial feasibility                                                                                                                      | 08 |
|   | (d) Describe in brief the important components of a project feasibility report.                                                                                                                 | 12 |

**Section – B**

- |   |                                                                                                                                                                                                                                                                                                                  |    |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 5 | (a) Define IRR. Write down the limitation of Net present value as investment criteria.                                                                                                                                                                                                                           | 10 |
|   | (b) Write short notes on the followings:<br>(i) Ordinary Annuity, (ii) Opportunity cost, (iii) Sources of risks of a project                                                                                                                                                                                     | 15 |
|   | (c) You are negotiating with the govt. The right to mine 150, 000 tons of iron ore per year for 18 years. The current price per ton of iron ore is tk. 3000 and it is expected to increase at the rate of 6% per year. What is the present value of the iron ore that you can mine if the discount rate is 16% ? | 10 |
| 6 | (a) “ A taka today is more valuable than a taka a year hence.” – Justify the statement.                                                                                                                                                                                                                          | 10 |
|   | (b) Distinguish between pay-back period and discounted payback period with necessary examples.                                                                                                                                                                                                                   | 10 |
|   | (c) An initial investment of 2.5 million BDT is expected to generate 0.5 million BDT per year for 7 years. Calculate the discounted payback period of the investment if the discount rate is 10%.                                                                                                                | 15 |
| 7 | (a) Distinguish between equity share and preference share with example.                                                                                                                                                                                                                                          | 10 |

- (b) Discribe four important investment criteria with necessary examples. 15
- (c) Calculate the IRR of a city link road project from the following data: 10  
Lower discount rate = 11%  
Higher discount rate = 16%
- NPV at lower discount rate = 1.947  
NPV at higher discount rate = - 0.37
- 8 (a) What is cost of capital? How the time horizon of a cash flow estimate is determined? Explain. 10
- (b) Illustrate the types of cash flows of a project with examples. 10
- (c) What is meant by FVIFA? What are the appliation of future value annuity formula in our day to day life? Illustrate with necessary example. 15
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Khulna University of Engineering & Technology  
Department of Urban and Regional Planning  
BURP 4<sup>th</sup> Year 1<sup>st</sup> Term Examination, 2021  
URP 4171  
(Legal Basis of Planning)



Full Marks: 210

Time: 3.0 hrs

- N.B.** i) Answer any **three** questions from each section in separate script.  
ii) Figures in the right margin indicate full marks.

**Section – A**

1. (a) Discuss with examples the importance of studying Legal Basis of Planning for the students of Urban and Regional Planning. (15)  
(b) Define the following terms with examples. (20)
  1. Law
  2. Ordinance
  3. Regulations
  4. Act
  5. Rules.
2. (a) Make a list of clearance certificates and other requirements that need to submit with application (form) for the approval of a private housing project in Bangladesh. (10)  
(b) According to the Private Housing Project Land Development Rule 2004 (amended in 2012), what is the procedure for approval of a private housing project in Bangladesh? Elaborate in brief. (15)  
(c) Illustrate the requirements of lay out plan of a private housing project in Bangladesh. (10)
3. (a) Which factors are responsible for the lengthy process of land acquisition and compensation? Suggest some measures to solve the related problems of land acquisition and compensation. (15)  
(b) Analyze the steps of building plan approval process according to the Bangladesh Building Construction Rules 1996. (20)
4. (a) Describe the provision laid down in the Town Improvement Act 1953 as regards Improvement scheme and Re-housing scheme. (10)  
(b) What are the provisions kept in Environmental Conservation Rule 1997 (amended in 2002) to declare an area as Ecologically Critical Area. Illustrate in brief. (10)  
(c) Describe the key elements of Playfield, Open space, Park, Garden and Natural Tank/Pond in Urban Areas Conservation Act 2000. (15)

**Section – B**

5. (a) Town Improvement Act was the first legal basis for planned urbanization and development in Bangladesh. Evaluate this statement. (15)  
(b) Discuss the governance structure of DIT. (10)  
(c) Why and when RAJUK was emerged? (10)
6. (a) Why is a unique building code important for a country? (05)  
(b) Evaluate the planning aspects considered in the updated BNBC. (20)  
(c) Analyze the causes and remedies of deviations in building construction in the urban areas of Bangladesh. (10)
7. (a) Execution of proposed URP Act 2017 would require coordinated efforts of many ministries and departments. Mention them with the reasons. (15)  
(b) Narrate the functions of Urban and Regional Planning Executive Council under the draft URP Act 2017. (10)  
(c) Explain the importance of Planning Inspectors in comparison with Building Inspectors in Bangladesh. (10)
8. Write short notes from the following (any five) (35)
  - a) Building height regulation in Dhaka City
  - b) Experts and professionals engaged in updating BNBC
  - c) Sources of fund for a municipality
  - d) Declaration of city area
  - e) Ward Committee
  - f) Pourashava Master Plan
  - g) Bangladesh Building Regulatory Authority.